Sequences & Series

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Reading a sequence from its terms

The bar chart shows the first five terms of a sequence.

a Determine whether the sequence is arithmetic or geometric, and find the common ratio or difference. Geometric — each term is double the previous one, so $r = 2$.

b Find the 8th term. $a_8 = 2 \times 2^7 = 2 \times 128 = 256$.

(a) Geometric — each term is double the previous one, so $r = 2$.; (b) $a_8 = 2 \times 2^7 = 2 \times 128 = 256$.

2 [Foundational] • Target: Arithmetic sequences: finding terms

Find the 12th term of the arithmetic sequence with $a = 5$, $d = 3$.

$a_{12} = 5 + 11(3) = 5 + 33 = 38$.

3 [Foundational] • Target: Arithmetic sequences: finding terms

Find the 20th term of the arithmetic sequence with $a = -4$, $d = 2$.

$a_{20} = -4 + 19(2) = -4 + 38 = 34$.

4 [Foundational] • Target: Arithmetic sequences: finding terms

Find the 15th term of the arithmetic sequence with $a = 100$, $d = -6$.

$a_{15} = 100 + 14(-6) = 100 - 84 = 16$.

5 [Foundational] • Target: Arithmetic sequences: finding terms

Write the first four terms of the arithmetic sequence with $a = 7$, $d = -2$.

$7,\ 5,\ 3,\ 1$.

6 [Foundational] • Target: Arithmetic sequences: finding a and d

The 4th term of an arithmetic sequence is 17 and the 9th term is 42. Find the first term a and common difference d.

$a + 3d = 17$ and $a + 8d = 42$. Subtracting: $5d = 25 \Rightarrow d = 5$. Then $a = 17 - 15 = 2$.

7 [Foundational] • Target: Arithmetic sequences: finding a and d

An arithmetic sequence has $a_1 = 10$ and $a_6 = 30$. Find the common difference d.

$a_6 = a_1 + 5d = 10 + 5d = 30 \Rightarrow 5d = 20 \Rightarrow d = 4$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Arithmetic sequences: finding a and d

An arithmetic sequence has $a_3 = 11$ and $a_7 = 27$. Find the common difference d.

$a_7 - a_3 = 4d = 16 \Rightarrow d = 4$.

9 [Intermediate] • Target: Arithmetic series: the sum formula

Find the sum of the first 20 terms of the arithmetic sequence with $a = 3$, $d = 4$, using $S_n = \frac{n}{2}(2a + (n-1)d)$.

$S_{20} = \frac{20}{2}(6 + 19(4)) = 10(6 + 76) = 10(82) = 820$.

10 [Intermediate] • Target: Arithmetic series: the sum formula

Find the sum of the first 15 terms of the arithmetic sequence with $a = 50$, $d = -3$.

$S_{15} = \frac{15}{2}(100 + 14(-3)) = 7.5(100 - 42) = 7.5(58) = 435$.

11 [Intermediate] • Target: Arithmetic series: the sum formula

Find the sum of the arithmetic series $2 + 5 + 8 + \cdots + 50$.

Here $a = 2$, $d = 3$; $50 = 2 + (n-1)3 \Rightarrow n = 17$. $S_{17} = \frac{17}{2}(2 + 50) = 17(26) = 442$.

12 [Intermediate] • Target: Arithmetic series: the sum formula

Find the sum of the first 10 positive multiples of 6.

These form an arithmetic sequence with $a = 6$, $d = 6$, $n = 10$.
$S_{10} = \frac{10}{2}(12 + 9(6)) = 5(12 + 54) = 5(66) = 330$.

13 [Intermediate] • Target: Geometric sequences: finding terms

Find the 6th term of the geometric sequence with $a = 3$, $r = 2$.

$a_6 = 3(2)^5 = 3(32) = 96$.

14 [Intermediate] • Target: Geometric sequences: finding terms

Find the 5th term of the geometric sequence with $a = 100$, $r = 0.5$.

$a_5 = 100(0.5)^4 = 100(0.0625) = 6.25$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Geometric sequences: finding terms • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

Find the 4th term of the geometric sequence with $a = 2$, $r = -3$.

$a_4 = 2(-3)^3 = 2(-27) = -54$.

16 [Challenging] • Target: Geometric sequences: finding terms • Cross-Lesson: 6.01 Limits & Derivatives

Write the first four terms of the geometric sequence with $a = 5$, $r = 3$.

$5,\ 15,\ 45,\ 135$.

17 [Challenging] • Target: Geometric sequences: finding a and r • Cross-Lesson: 7.01 Integration & Applications

The 2nd term of a geometric sequence is 6 and the 5th term is 162. Find r and a.

$ar = 6$ and $ar^4 = 162$. Dividing: $r^3 = 27 \Rightarrow r = 3$. Then $a = 6/3 = 2$.

18 [Challenging] • Target: Geometric sequences: finding a and r • Cross-Lesson: 8.01 Exponentials & Logarithms

A geometric sequence has $a_1 = 4$ and $a_4 = 108$. Find r.

$a_4 = a_1 r^3 = 4r^3 = 108 \Rightarrow r^3 = 27 \Rightarrow r = 3$.

19 [Challenging] • Target: Geometric sequences: finding a and r • Cross-Lesson: 9.01 Statistics & Probability

A geometric sequence has 3rd term 12 and 6th term 96. Find a and r.

$ar^2 = 12$ and $ar^5 = 96$. Dividing: $r^3 = 8 \Rightarrow r = 2$. Then $a = 12/4 = 3$.

20 [Challenging] • Target: Geometric series: the finite sum • Cross-Lesson: 1.01 Algebra & Equations

Find the sum of the first 6 terms of the geometric series with $a = 3$, $r = 2$, using $S_n = \frac{a(r^n - 1)}{r - 1}$.

$S_6 = \frac{3(2^6 - 1)}{2 - 1} = 3(63) = 189$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: MCQ — Reading a sequence from its terms

The bar chart shows the first five terms of a sequence. Find the common difference.

a. 6

b. 5 ✓

c. 4

d. 9

Each term increases by 5.

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22 [Foundational] • Target: MCQ — Reading a sequence from its terms

Using the same bar chart, predict the 8th term.

a. 29

b. 34

c. 39 ✓

d. 44

$d = 5$, $a = 4$: $T_8 = 4 + 7(5) = 39$.

23 [Foundational] • Target: MCQ — Reading a sequence from its terms

Using the same bar chart, find the sum of the first five terms shown.

a. 70 ✓

b. 66

c. 24

d. 75

$4 + 9 + 14 + 19 + 24 = 70$.

24 [Foundational] • Target: MCQ — Arithmetic sequences: finding terms

Find the 14th term of the arithmetic sequence with $a = 6$, $d = 4$.

a. 62

b. 54

c. 50

d. 58 ✓

$T_{14} = 6 + 13(4) = 58$.

25 [Foundational] • Target: MCQ — Arithmetic sequences: finding terms

Find the 25th term of the arithmetic sequence with $a = 2$, $d = 7$.

a. 177

b. 170 ✓

c. 175

d. 163

$T_{25} = 2 + 24(7) = 170$.

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26 [Foundational] • Target: MCQ — Arithmetic sequences: finding a and d

The 3rd term of an arithmetic sequence is 15 and the 8th term is 45. Find the first term a and common difference d.

a. $a = 3, d = 5$

b. $a = 3, d = 6$ ✓

c. $a = 6, d = 3$

d. $a = 9, d = 6$

$d = \frac{45 - 15}{5} = 6$; $a = 15 - 2(6) = 3$.

27 [Foundational] • Target: MCQ — Arithmetic series: the sum formula

Find the sum of the first 30 terms of the arithmetic sequence with $a = 1$, $d = 2$ (the odd numbers).

a. 960

b. 930

c. 870

d. 900 ✓

$S_{30} = \frac{30}{2}(2 + 29(2)) = 15(60) = 900$.

28 [Foundational] • Target: MCQ — Geometric sequences: finding terms

Write the first four terms of the geometric sequence with $a = 4$, $r = 2$.

a. $4, 6, 8, 10$

b. $4, 16, 32, 64$

c. $4, 8, 12, 16$

d. $4, 8, 16, 32$ ✓

Each term doubles: $4, 8, 16, 32$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: MCQ — Geometric sequences: finding a and r

A geometric sequence has 2nd term 4 and 5th term -108. Find r.

a. $r = 3$

b. $r = -27$

c. $r = -3$ ✓

d. $r = -9$

$r^3 = -108 \div 4 = -27 \Rightarrow r = -3$.

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30 [Intermediate] • Target: MCQ — Geometric series: the finite sum

Find the sum of the first 8 terms of the geometric series with $a = 1$, $r = 2$.

a. 255 ✓

b. 128

c. 254

d. 256

$S_8 = \frac{1(2^8 - 1)}{1} = 255$.

31 [Intermediate] • Target: MCQ — Infinite geometric series

Express the repeating decimal $0.777\ldots$ as a fraction using an infinite geometric series with $a = 0.7$, $r = 0.1$.

a. $\frac{7}{90}$

b. $\frac{77}{100}$

c. $\frac{7}{9}$ ✓

d. $\frac{7}{10}$

$S = \frac{0.7}{1 - 0.1} = \frac{0.7}{0.9} = \frac{7}{9}$.

32 [Intermediate] • Target: MCQ — Sigma notation

Use $\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$ to evaluate $\sum_{k=1}^{15} k$.

a. 120 ✓

b. 110

c. 105

d. 136

$\frac{15(16)}{2} = 120$.

33 [Intermediate] • Target: MCQ — Recursive sequences

A sequence satisfies $a_1 = 2$ and $a_{n+1} = 2a_n + 1$. Find a_4.

a. 11

b. 23 ✓

c. 5

d. 47

$2, 5, 11, 23$: $a_4 = 23$.

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34 [Intermediate] • Target: MCQ — Identifying arithmetic and geometric sequences

State whether $5, 10, 20, 40, \ldots$ is arithmetic or geometric, and give the common difference or ratio.

a. Geometric, $r = 2$ ✓

b. Geometric, $r = 5$

c. Arithmetic, $d = 2$

d. Arithmetic, $d = 5$

Each term is doubled: geometric, $r = 2$.

35 [Intermediate] • Target: MCQ — Arithmetic series: the sum formula

Find the 16th term of the arithmetic sequence with $a = 20$, $d = -7$.

a. -82

b. -92

c. -85 ✓

d. -78

$a_{16} = 20 + 15((-7)) = 20 - 105 = -85$.

36 [Intermediate] • Target: MCQ — Geometric sequences: finding terms

Find the 13th term of the arithmetic sequence with $a = -16$, $d = 9$.

a. 95

b. 101

c. 83

d. 92 ✓

$a_{13} = -16 + 12(9) = -16 + 108 = 92$.

37 [Intermediate] • Target: MCQ — Geometric sequences: finding a and r

Find the 8th term of the arithmetic sequence with $a = 6$, $d = 2$.

a. 23

b. 22

c. 18

d. 20 ✓

$a_8 = 6 + 7(2) = 6 + 14 = 20$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: MCQ — Geometric series: the finite sum • Cross-Lesson: 8.01 Exponentials & Logarithms

Find the 5th term of the arithmetic sequence with $a = -2$, $d = 9$.

a. 25

b. 43

c. 37

d. 34 ✓

$a_5 = -2 + 4(9) = -2 + 36 = 34$.

39 [Challenging] • Target: MCQ — Infinite geometric series • Cross-Lesson: 9.01 Statistics & Probability

Find the 6th term of the geometric sequence with $a = -9$, $r = -2$.

a. 291

b. -288

c. 285

d. 288 ✓

$a_6 = -9((-2))^5 = -9((-32)) = 288$.

40 [Challenging] • Target: MCQ — Sigma notation • Cross-Lesson: 1.01 Algebra & Equations

Find the 4th term of the geometric sequence with $a = 1$, $r = -3$.

a. -27 ✓

b. -30

c. 27

d. -24

$a_4 = 1((-3))^3 = 1((-27)) = -27$.

41 [Challenging] • Target: MCQ — Recursive sequences • Cross-Lesson: 2.01 Functions & Transformations

Find the 7th term of the geometric sequence with $a = -7$, $r = -2$.

a. 448

b. -445

c. -448 ✓

d. -451

$a_7 = -7((-2))^6 = -7(64) = -448$.

42 [Challenging] • Target: MCQ — Identifying arithmetic and geometric sequences • Cross-Lesson: 4.01 Trigonometry

Find the sum of the infinite geometric series with $a = 16$, $r = \frac{1}{4}$.

a. $\frac{64}{3}$ ✓

b. $-\frac{64}{3}$

c. $\frac{58}{3}$

d. $\frac{70}{3}$

$1 - r = 1 - \frac{1}{4} = \frac{3}{4}$. $S = \frac{a}{1-r} = 16 \times \frac{4}{3} = \frac{64}{3}$.

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43 [Challenging] • Target: MCQ — Reading a sequence from its terms • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

Evaluate $\sum_{k=1}^{6}(2k+2)$.

a. -54

b. 54 ✓

c. 50

d. 58

Terms: $4, 6, 8, 10, 12, 14$. Sum $= 54$.

44 [Challenging] • Target: MCQ — Word problems: sequences in context • Cross-Lesson: 6.01 Limits & Derivatives

A sequence satisfies $a_1 = 2$ and $a_{n+1} = a_n + 4$. Find a_4.

a. 10

b. 14 ✓

c. 17

d. 11

$a_1 = 2, a_2 = 6, a_3 = 10, a_4 = 14$.

45 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 7.01 Integration & Applications

Find the sum of the first 7 terms of the geometric series with $a = 2$, $r = 2$, using $S_n = \frac{a(r^n - 1)}{r - 1}$.

a. 256

b. 508

c. 254 ✓

d. 127

$S_7 = \frac{2(2^7 - 1)}{1} = 2(127) = 254$.

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